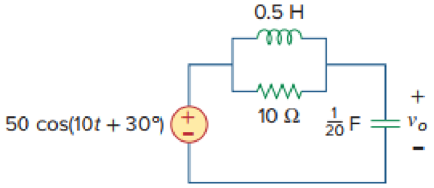
Chapter 9, Problem 11PP

Problem

Calculate *vo* in the circuit of Fig. 9.27.

Figure 9.27:



Step-by-step solution

Step 1 of 8

Find the phasor corresponding to the sinusoid $v(t)=50\cos(10t+30^\circ)$ V.

$$v(t)=50\cos(10t+30^\circ)\ \text{V}$$
$$=50\angle 30^\circ\ \text{V}$$

Therefore, the phasor $\mathbf{V}_s$ corresponding to sinusoid $v(t)=50\cos(10t+30^\circ)$ V is $50\angle 30^\circ$ V.

Calculate the value of the impedance $\mathbf{Z}_C$ of the $\frac{1}{20}$ F capacitor at $\omega=10$ rad/s.

Write the expression for the impedance $\mathbf{Z}_C$ of the $\frac{1}{20}$ F capacitor at $\omega=10$ rad/s.

$$\mathbf{Z}_C=\frac{-j}{\omega C}$$
$$\mathbf{Z}_C=\frac{-j}{10\left(\frac{1}{20}\right)}$$
$$=-j2$$

Therefore, the value of impedance $\mathbf{Z}_C$ of the $\frac{1}{20}$ F capacitor at $\omega=10$ rad/s is $-j2\ \Omega$.

Step 2 of 8

Calculate the value of the impedance $\mathbf{Z}_L$ of the 0.5 H inductor at $\omega=10$ rad/s.

Write the expression for the impedance $\mathbf{Z}_L$ of the 0.5 H inductor r at $\omega=10$ rad/s.

$$\mathbf{Z}_L=j\omega L$$
$$\mathbf{Z}_L=j10(0.5)$$
$$=j5$$

Therefore, the value of impedance $\mathbf{Z}_L$ of the 0.5 H inductor at $\omega=10$ rad/s is $j5\ \Omega$.

Step 3 of 8

Draw the phasor domain equivalent circuit of Figure 1 as shown in Figure 2.

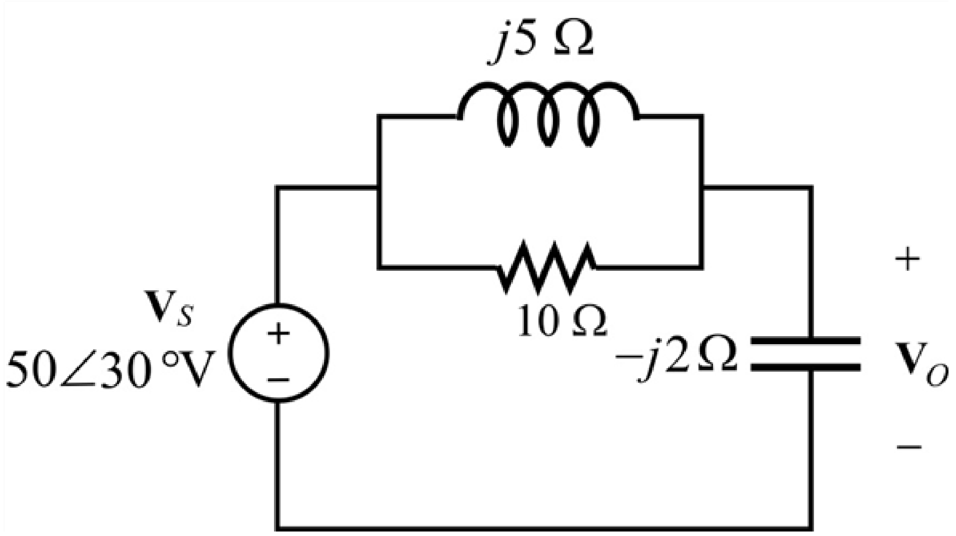


Figure 2: Phasor domain equivalent circuit.

Step 4 of 8

Consider $\mathbf{Z}_1$ is the impedance of the $0.5\ \mathrm{H}$ inductor in parallel with the resistor $10\ \Omega$.

$\mathbf{Z}_2$ is the impedance of the $\frac{1}{20}\ \mathrm{F}$ capacitor.

Redraw the Figure 1 as shown in Figure 2 for analysis.

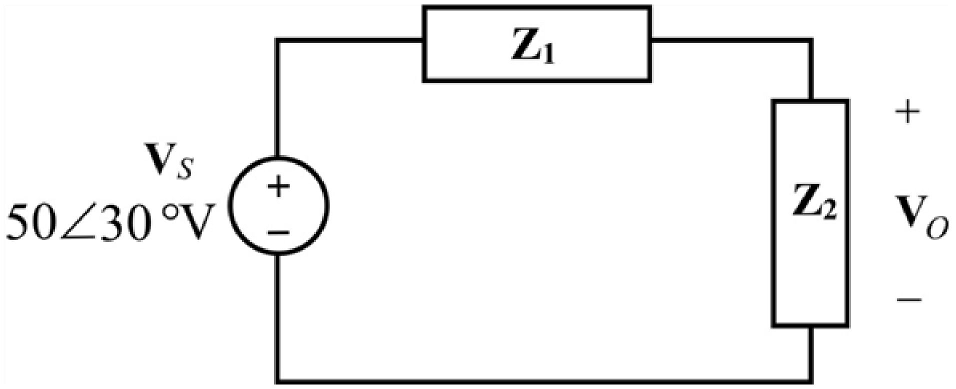


Figure 3: Simplified Phasor domain equivalent circuit.

Step 5 of 8

Calculate the value of $\mathbf{Z}_1$.

Write the expression for $\mathbf{Z}_1$.

$$\begin{aligned}\mathbf{Z}_1 &= \mathbf{Z}_L \parallel 10 \\ &= \frac{\mathbf{Z}_L(10)}{\mathbf{Z}_L + 10}\end{aligned}$$

Substitute $j5\ \Omega$ for $\mathbf{Z}_L$ in this expression.

$$\begin{aligned}\mathbf{Z}_1 &= \frac{j5(10)}{j5+10} \\ &= \frac{j50}{j5+10} \\ &= \left(\frac{j10}{2+j}\right) \\ &= (2+j4)\ \Omega\end{aligned}$$

Therefore, the value of $\mathbf{Z}_1$ is $(2+j4)\ \Omega$.

Step 6 of 8

Calculate the value of the output voltage $\mathbf{V}_o$.

Apply voltage division rule across the impedance $\mathbf{Z}_2$ to find the output voltage $\mathbf{V}_o$.

$$\mathbf{V}_o = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}_s$$

Substitute $(2+j4)\ \Omega$ for $\mathbf{Z}_1$, $-j2\ \Omega$ for $\mathbf{Z}_2$ in this expression.

$$\begin{aligned}\mathbf{V}_o &= \frac{-j2}{2+j4-j2}50\angle 30^\circ \\ &= \frac{-j2}{2+j2}50\angle 30^\circ \\ &= \frac{-j}{1+j}50\angle 30^\circ \\ &= (-0.5-j0.5)50\angle 30^\circ\end{aligned}$$

Step 7 of 8

Simplify $\mathbf{V}_o$ further.

$$\mathbf{V}_o = 35.36\angle -105^\circ$$

Therefore, the value of the output voltage $\mathbf{V}_o$ in phasor form is $35.36\angle -105^\circ\ \text{V}$.

Step 8 of 8

Find the sinusoid corresponding to the phasor output voltage $\mathbf{V}_o = 35.36\angle -105^\circ\ \text{V}$

$$\begin{aligned}\mathbf{V}_o &= 35.36\angle -105^\circ\ \text{V} \\ v_o(t) &= 35.36\cos(10t - 105^\circ)\ \text{V}\end{aligned}$$

Therefore, the value of the required output voltage $v_o(t)$ is $\boxed{35.36\cos(10t-105^\circ)\ \text{V}}$.